Art Gallery Problems for Orthogonal Polygons and Orthogonal Polyhedra

A Literature Review of Results and Open Problems

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Abstract

This review surveys art-gallery problems for orthogonal polygons, polyominoes, and orthogonal polyhedra, with emphasis on perimeter-type bounds, holes, and the model changes that produce current open problems. In two dimensions, the perimeter-over-six question is meaningful only after fixing a lattice scale. Maßberg proves the tight $\lfloor \ell/6 \rfloor$ point-guard bound for hole-free polyominoes, and that theorem also covers hole-free integral orthogonal polygons after unit-grid subdivision. The unresolved perimeter-over-six direction is therefore the version with holes, or a broader lattice-domain formulation not reducible to the hole-free polyomino case. In discrete 3D and higher dimensions, the cell-volume analogue is known: Pinciu proves the tight $\lfloor (m+1)/3 \rfloor$ point-guard theorem for m-polyhypercubes. For arbitrary continuous orthogonal polyhedra, however, point guards do not give a linear planar analogue: some examples require $\Theta(n^{3/2})$ guards. The continuous 3D literature consequently turns to edge guards, reflex-edge guards, and face guards, while discrete voxel/cell guard variants form a separate line of open problems.

Contents

1 Executive Synthesis

The review has one organizing question: which art-gallery bounds survive when the object is orthogonal, the perimeter is lattice-normalized, holes may be present, and the dimension changes? The answer is not a single theorem but a map of comparable models. The sharp perimeter-over-six theorem is known for hole-free polyominoes, and the same theorem settles the hole-free integral orthogonal-polygon reading by subdivision. The remaining two-dimensional frontier is the holes version, together with any broader lattice-domain formulation that does not reduce to a hole-free polyomino. In discrete 3D and higher dimensions, Pinciu gives the direct point-guard cell-volume theorem for polyhypercubes. In continuous orthogonal polyhedra, the direct point-guard analogy fails, and edge, reflex-edge, face, and cell-guard variants carry the relevant open questions.

Result line	Current status	Why it matters here	Main evidence
Simple orthogonal polygons	Tight $\lfloor n/4 \rfloor$ vertex/point-guard baseline for hole-free orthogonal polygons.	Establishes the classical 2D orthogonal benchmark, but it is a vertex-count theorem, not a perimeter theorem.	Kahn–Klawe–Kleitman; Gyori [16, 14]
Hole-free polyomino perimeter	Tight $\max\{1, \lfloor \ell/6 \rfloor\}$ point-guard theorem.	This is the core perimeter-over-six result and the source of the hole-free integral-polygon overlap.	Maßberg [19, Theorem 1]
Integral orthogonal polygons, no holes	Solved by reduction to hole-free polyominoes after unit-grid subdivision.	Prevents the Diaz-Banez et al. $\lfloor N/6 \rfloor$ question from being misread as open in the hole-free case.	Maßberg; Diaz-Banez et al. [19, Theorem 1]; [9]
Polyomino or integral domains with holes	Open or conditional for perimeter-over-six point guards.	This is the main 2D open direction closest to the original perimeter question.	Maßberg habilitation [20, Conj. 6.10]
Orthogonal polygons with holes, vertex guards	Partially solved; strongest conjectured bounds remain open.	These results explain the role of holes, but they do not settle point-guard perimeter bounds.	Urrutia; Zylinski; Michael–Pinciu [29, 34, 21]
Polycubes and polyhypercubes	Tight $\lfloor (m+1)/3 \rfloor$ point-guard cell-volume theorem.	This is the genuine 3D-and-higher analogue of the polyomino cell-count theorem, but only for unit-cell complexes.	Pinciu [25]
Continuous orthogonal polyhedra in 3D	No direct point-guard analogue; edge and reflex-edge variants remain active.	The continuous 3D branch must be read through guard model changes rather than by copying the planar perimeter theorem.	Paterson–Yao; Viglietta; Benbernou et al.; Cano et al. [24, 30, 3, 7]

Table 1: *Reader-facing synthesis of the central result lines. Each row is intended to be read only under the stated model; later sections give the definitions, non-implications, and evidence trail.*

The document is assembled in layers. The front matter gives the short answer; the terminology section fixes the model axes; the historical flow explains how the result lines diverged; the status matrix and coverage chart show what is proved, conditional, or open; the non-implication and proof-technique sections explain why adjacent theorems do not automatically transfer; the model-by-model sections give the detailed survey; and the open-problem, source-guide, and evidence sections make the unresolved directions auditable.

2 Introduction and Scope

The art-gallery literature is best read through four modeling choices: the object class, the presence of holes, the allowed guard model, and the parameter in the bound. A statement about point guards in a hole-free polyomino need not imply a statement about vertex guards in a polygon with holes, and neither statement transfers automatically to edge guards in an orthogonal polyhedron. This review keeps those distinctions explicit because most apparent conflicts among the results come from changing one of these choices.

For two-dimensional orthogonal objects, the main perimeter issue is scale. There is no scale-free theorem of the form "at most $P/6$ guards" for arbitrary real-coordinate orthogonal polygons when P is Euclidean perimeter: scaling the polygon changes P without changing the guard number. Perimeter-over-six results therefore belong to lattice-normalized settings, such as polyomino perimeter, integral perimeter, or the unit-edge ortho-unit model.

Main 2D Distinction

If Q is a hole-free integral orthogonal polygon of lattice perimeter N, unit-grid subdivision identifies it with a hole-free polyomino of perimeter $\ell = N$. Maßberg's theorem then gives $\max\{1, \lfloor N/6 \rfloor\}$ point guards, and hence $\lfloor N/6 \rfloor$ guards whenever $N \geq 6$. Thus the nontrivial hole-free integral-polygon reading of the Diaz-Banez et al. $\lfloor N/6 \rfloor$ question is already covered by Maßberg; the genuinely unresolved reading is the extension to holes or to a broader lattice-domain convention.

In three dimensions the direct point-guard analogy breaks for a different reason. Orthogonal polyhedra can require superlinear numbers of point guards in the number of vertices, so the 3D branch of the literature usually asks for edge guards, reflex-edge guards, face guards, or guards in discrete polycube visibility models. The word *orthonormal* is not the relevant geometric term; the standard objects are orthogonal polygons, orthogonal polyhedra, and orthogonal polytopes.

The following table summarizes the readings of the perimeter-over-six phrase that recur in the literature.

Reading of the $\lfloor N/6 \rfloor$ question	Status	Reason
Simple/hole-free integral orthogonal polygon	Solved	Same object as a hole-free polyomino after unit-grid subdivision; $N = \ell$.
Ortho-unit polygon	Solved, stronger	Diaz-Banez et al. prove $\lfloor (n-4)/8 \rfloor$, and here $n = P = N$.
Integral or polyomino domain with holes	Open/conditional	Maßberg's hole extension depends on an unproved maximal-rectangle packing conjecture.
Real-coordinate orthogonal polygon with Euclidean perimeter	Not a valid scale-free statement	Uniform scaling changes perimeter but not guard number.

3 Conventions and Terminology

2D objects. An *orthogonal polygon* is a polygon whose edges are horizontal or vertical; the literature also uses *rectilinear* for the same class. A *polyomino* is a finite edge-connected union of unit axis-parallel squares; its perimeter is the number of unit boundary edges, including hole boundaries when holes are present. An *integral orthogonal polygon* has integer coordinates, so its perimeter is measured as a lattice length. An *ortho-unit polygon* is an orthogonal polygon all of whose edges have length one; in that model, the number of vertices equals the perimeter.

3D objects. The standard 3D term is *orthogonal polyhedron*: a polyhedron whose faces are axis-aligned, equivalently whose face normals are coordinate directions. In higher dimensions, the corresponding term is *orthogonal polytope*. The word *orthonormal* describes vectors or bases, not galleries. Polycubes are the voxel analogue of polyominoes and form a discrete subclass of orthogonal polyhedra. A polyhypercube is the corresponding finite face-connected union of unit hypercubes in dimension d.

Guard models. Unless a section says otherwise, a 2D guard is a standard point guard with straight-line visibility inside the polygon. In 3D the model must be stated: the literature distinguishes point guards, vertex guards, edge guards, reflex-edge guards, face guards, open edge guards, and discrete polycube visibility graph models. These models are not interchangeable.

Why perimeter needs a lattice scale. No theorem of the form "at most $P/6$ guards" can hold for arbitrary real-coordinate orthogonal polygons if P is Euclidean perimeter: scaling a polygon down preserves the guard number while making P arbitrarily small. The perimeter-over-six question is therefore a lattice-normalized question: polyomino perimeter, integral perimeter, or a fixed unit-edge model.

Common parameters. The same letters are used differently across the literature, so the review uses the following conventions unless a cited theorem uses its own notation.

Symbol	Meaning in this review	Main caution
n	Number of polygon vertices, or polyhedron vertices when the 3D context is explicit.	Vertex-count bounds are not perimeter bounds.
h	Number of holes in a 2D polygonal domain.	Holes can change both decomposition structure and guard type.
r	Number of reflex edges in a 3D orthogonal polyhedron.	This is a 3D parameter; in 2D, reflex vertices play the analogous role.
m	Number of polyomino cells in 2D, or number of polyhedron edges in 3D when stated.	Area/cell bounds and edge-count bounds should not be compared without the model.
ℓ	Lattice perimeter of a polyomino, including hole boundaries.	This is the parameter in Maßberg's perimeter theorem.
N	Lattice perimeter of an integral orthogonal polygon.	After unit-grid subdivision, $N = \ell$ for the corresponding polyomino.
P	Euclidean perimeter when discussing scale, or a polygon only when no formula is involved.	Real-coordinate Euclidean perimeter is not scale-free.

4 Historical Flow

1. Classical art-gallery theorems start with arbitrary simple polygons: Chvatal proves $\lfloor n/3 \rfloor$ and Fisk gives the standard triangulation proof [8, 12]. Shermer, O'Rourke, and Urrutia provide the main survey/book context [27, 23, 28].

2. Orthogonal polygons admit stronger vertex-count theorems: Kahn–Klawe–Kleitman prove the tight $\lfloor n/4 \rfloor$ bound for simple orthogonal polygons [16].

3. Holes complicate the 2D theory. Vertex-guard conjectures for orthogonal polygons with holes, especially Shermer's and Hoffmann's formulations, remain separate from the perimeter-over-six point-guard question [29, 15, 34, 21].

4. Polyominoes introduce area and lattice-perimeter parameters. The tight cell-count theorem covers connected m-polyominoes with $m \geq 2$, holes allowed [4, Cor. 1], while Maßberg's perimeter-over-six theorem is proven for hole-free polyominoes and becomes conditional in the presence of holes [19, Theorem 1]; [20, Thm. 6.3 and Conj. 6.10].

5. Recent ortho-unit work proves a strong unit-edge theorem and states an integral-perimeter conjecture [9]. Its literal hole-free integral-polygon reading overlaps with Maßberg's polyomino theorem; only the holes or broader-domain reading remains open.

6. In 3D, point guards are too weak for a planar-style theorem: orthogonal polyhedra need $\Theta(n^{3/2})$ point guards in the worst case [24, 30]. Edge-guard and reflex-edge-guard conjectures become the main 3D analogue [3, 32, 7].

5 Status Overview

Table 2 separates the main models discussed below. The labels "proven tight," "conditional," "conjectural," and "open" refer to the exact model in the first three columns; changing the guard type, allowing holes, or changing the parameter can change the answer.

Figure 1 summarizes which two-dimensional polygon classes are actually covered by the main bounds in this review. Green cells are proven coverage, pale green cells are proven but usually weaker than a later special theorem, blue cells indicate a proper-subclass or different-parameter theorem, yellow cells mark conditional or open status, and gray cells are outside the theorem as stated.

2D object class	Classical point $\lfloor n/3 \rfloor$	Orthogonal point $\lfloor n/4 \rfloor$	Orthogonal holes, vertex guards	Perimeter point $\lfloor \ell/6 \rfloor$ or $\lfloor N/6 \rfloor$	Area point $\lfloor (m+1)/3 \rfloor$	Ortho-unit point $\lfloor (n-4)/8 \rfloor$
General simple polygon, no holes	yes, tight	–	–	–	–	–
Simple orthogonal polygon, no holes	yes, weaker	yes, tight	–	–	–	–
Hole-free integral orthogonal polygon	yes, weaker	yes, vertex bound	–	yes, by subdivision	yes, area parameter	only unit-edge subclass
Hole-free polyomino	yes, weaker	yes, vertex bound	–	yes, tight	yes, area parameter	only unit-edge subclass
Ortho-unit polygon	yes, weaker	yes, vertex bound	–	yes, follows	yes, area parameter	yes, tight
Orthogonal domain with holes, real coordinates	–	–	partial/open	–	–	–
Integral or polyomino domain with holes	–	–	vertex-only partial/open	cond./open	yes, tight	–

Figure 1: *Two-dimensional coverage chart for the principal bounds. A dash means the theorem does not apply as stated, usually because the object has holes, lacks a lattice scale, uses the wrong guard model, or is controlled by a different parameter.*

Model	Guards	Parameter	Status	Main source
Simple orthogonal polygon	point	vertices n	Proven tight: $\lfloor n/4 \rfloor$	Kahn–Klawe–Kleitman; Gyori [16, 14]
Orthogonal polygon with holes	vertex	vertices n, holes h	Open at the strongest conjectured bounds; improved h-independent upper bounds and partial sharp cases are known	Urrutia; Hoffmann–Kriegel; Zylinski; Michael–Pinciu [29, 15, 34, 21]
Hole-free polyomino	point	lattice perimeter ℓ	Proven tight for $\ell \geq 6$: $\lfloor \ell/6 \rfloor$	Maßberg [19, Theorem 1]
Polyomino with holes	point	lattice perimeter ℓ	Conditional/open: Maßberg's extension depends on a rectangle-packing conjecture	Maßberg habilitation [20, Conj. 6.10]
Polyomino with holes	point	cells m	Proven tight for $m \geq 2$: $\lfloor (m+1)/3 \rfloor$; $m = 1$ is trivial; does not imply perimeter-over-six	Biedl et al. [4, Cor. 1]
Polyhypercube	point	cells m	Proven tight for $m \geq 2$ and $d \geq 2$: $\lfloor (m+1)/3 \rfloor$; this is the cell-volume analogue, not a continuous polyhedron theorem	Pinciu [25]
Polyhypercube	voxel/cell	cells m, dimension d	Open with partial bounds: Pinciu gives lower and upper bounds and conjectures a sharp dependent-on-d formula	Pinciu [25]
Ortho-unit polygon	point	perimeter $P = n$	Proven tight for $n \geq 12$: $\lfloor (n-4)/8 \rfloor$, stronger than $P/6$ in this model	Diaz-Banez et al. [9]
Integral orthogonal polygon, hole-free	point	lattice perimeter N	Solved by overlap: subdivision gives a hole-free polyomino with $N = \ell$, so Maßberg proves the nontrivial Diaz-Banez $\lfloor N/6 \rfloor$ bound in this case	Maßberg; Diaz-Banez et al. [19, Theorem 1]; [9]
Integral or polyomino domain with holes	point	lattice perimeter N or ℓ	Still open/conditional: Maßberg's hole extension depends on a rectangle-packing conjecture	Maßberg habilitation [20, Conj. 6.10]
Orthogonal polyhedron	point	vertices n	No linear planar analogue: worst case is $\Theta(n^{3/2})$	Paterson–Yao; Viglietta [24, 30]
Orthogonal polyhedron	edge or reflex edge	edges m, reflex edges r	Main 3D analogue; Urrutia-type conjectures remain open beyond partial classes	Benbernou et al.; Viglietta; Cano et al. [3, 32, 7]

Table 2: *Status map for orthogonal art-gallery problems closest to perimeter-over-six.*

6 Model Separations and Non-Implications

The most useful way to test whether a proposed theorem is genuinely new is to ask which model changes are being made. The following separations account for most apparent overlaps in this part of the literature.

Scale separation.
A Euclidean perimeter statement for arbitrary real-coordinate orthogonal polygons cannot be scale-free. Any perimeter-over-six theorem must specify a unit: polyomino boundary length, integral lattice length, or unit-edge ortho-unit length.

Point guards versus vertex guards.
Vertex-guard theorems for orthogonal polygons with holes do not imply point-guard perimeter theorems. Conversely, a point-guard perimeter theorem need not place guards on vertices. This is why Shermer's and Hoffmann's vertex-guard conjectures are adjacent to, but distinct from, the polyomino perimeter question.

Standard visibility versus restricted visibility.
Floodlights, half-plane guards, sliding cameras, transmitters, k-hop visibility, and rook/queen polycube models are adjacent to standard point-guard or vertex-guard visibility, but they are not interchangeable with it. They can supply techniques and open problems, but each requires a separate model cell before its theorems can be used in this review. The current evidence matrix has separate cells for the currently cited sliding-camera, sliding-transmitter, k-hop, polyhypercube point-guard, polyhypercube pixel/voxel-guard, face-guard, and 2-reflex variants. It also has explicit scope-exclusion cells for floodlight, half-plane, dispersive, contiguous, mobile, point-boundary, and related variants that have not been promoted into theorem-level discussion.

Holes versus hole-free domains.
The hole-free perimeter theorem is not automatically stable under adding holes. Holes change the intersection graph of maximal rectangles and break the perfect-graph route used in the hole-free proof. Maßberg's maximal-rectangle packing conjecture is precisely the missing replacement structure in that setting.

Area versus perimeter.
The tight $\lfloor (m+1)/3 \rfloor$ theorem for connected m-cell polyominoes with holes, $m \geq 2$, is an area theorem; the one-cell case is the trivial one-guard case. It does not imply a perimeter theorem because polyominoes can have very different area-to-perimeter ratios, and the conjectured perimeter bound is sensitive to boundary complexity rather than cell count.

Ortho-unit versus integral.
Ortho-unit polygons form a narrow unit-edge subclass: every edge has length one, so vertex count and perimeter coincide. Integral orthogonal polygons allow longer lattice edges. In the hole-free case, subdivision reduces the integral model to hole-free polyominoes; with holes, it reduces to the polyomino-with-holes case, which is still conditional/open for perimeter.

2D point guards versus 3D point guards.
Orthogonal polyhedra do not inherit planar point-guard behavior. The $\Theta(n^{3/2})$ point-guard bound in 3D forces the analogue to move toward edge guards, reflex-edge guards, face guards, or discrete polycube models.

7 Proof Techniques and Where They Break

Triangulation and coloring.

Chvatal and Fisk establish the classical template for simple polygons [8, 12]. Kahn–Klawe–Kleitman and Gyori adapt coloring ideas to simple orthogonal polygons [16, 14]. The method is powerful for vertex-count bounds but does not directly produce a lattice-perimeter theorem.

Quadrilateralization and cactus duals.

Orthogonal holes are often attacked by decomposing the domain into rectangles or quadrilaterals and coloring a dual structure. Zylinski's cactus-dual condition explains why the $\lfloor (n+h)/4 \rfloor$ conjecture is approachable in some cases but not yet resolved in full [34]. Michael and Pinciu introduce same-sign diagonal graphs of convex quadrangulations and use vertex-cover bounds to unify several orthogonal-guarding results and improve h-independent bounds for polygons with holes [21].

Rectangle packing and perfect-graph methods.

Motwani–Raghunathan–Saran, Worman–Keil, and Maßberg connect orthogonal guarding with star polygons, rectangle systems, and perfect graphs [22, 33, 19]. This is the proof technology behind the hole-free polyomino perimeter theorem. Its failure point is explicit: the hole case depends on Maßberg's maximal-rectangle packing conjecture [20, Conj. 6.10].

Area and grid-cell arguments.

Biedl et al. prove the tight $\lfloor (m+1)/3 \rfloor$ theorem for connected m-polyominoes with $m \geq 2$ by using the cell structure rather than perimeter; the sufficiency proof works even for restrictive r-visibility [4, Theorem 1 and Cor. 1]. This is strong evidence that holes are tractable in a discrete model, but it leaves open whether perimeter alone controls the guard number.

Hardness and approximation reductions.

Schuchardt–Hecker, Katz–Roisman, and later work on sliding cameras and transmitters show that optimization versions remain hard even in restricted orthogonal settings [26, 17, 10, 5]. These results do not contradict extremal upper bounds; they warn that finding a minimum guard set is a different problem from proving a universal bound.

3D decomposition and binary-space partitions.

Paterson–Yao's binary-space-partition theorem, as used in Viglietta's survey, shows that point guards in orthogonal polyhedra have $\Theta(n^{3/2})$ worst-case behavior [24, 30]. This is the main reason a 3D analogue should not simply ask for a point-guard version of the planar perimeter theorem.

Edge-guard and reflex-edge induction.

Benbernou et al., Viglietta, and Cano–Toth–Urrutia–Viglietta show that the most promising 3D analogue is based on edges, reflex edges, and genus rather than point guards [3, 32, 7]. The remaining gap is structural: the known upper bounds do not meet the Urrutia-type lower-bound constructions.

8 Proof Roadmap for Core Result Lines

This section records what each main proof line actually buys. It is included to make the survey useful for readers who want to enter an open problem rather than only know its status.

8.1 Coloring Proofs in the Plane

The Chvatal–Fisk line proves existence by decomposing a polygon into simple pieces, coloring a graph derived from that decomposition, and choosing the smallest color class [8, 12]. The orthogonal Kahn–Klawe–Kleitman theorem uses the extra parity and degree structure of rectilinear polygons to improve the general $\lfloor n/3 \rfloor$

bound to $\lfloor n/4 \rfloor$ [16]; Gyori's proof is the shortest entry point for this result [14]. The transferable idea is the small-color-class argument. The non-transferable part is the parameter: these proofs count vertices, not lattice perimeter.

8.2 Holes and Quadrilateralization

For orthogonal polygons with holes, the natural decomposition is a quadrilateralization rather than a triangulation. The dual structure can fail to be a tree, so the coloring argument needs additional hypotheses. Zylinski isolates one such hypothesis through cactus-dual quadrilateralizations and recovers Aggarwal's theorem for $h \leq 2$ holes [34]. Michael–Pinciu give a different organizing language: same-sign diagonal graphs and vertex covers [21]. These papers show that holes are not merely a nuisance parameter; they alter the combinatorial object being colored or covered.

This is why the hole literature should be cited carefully. The results are highly relevant to orthogonal domains with holes, but their strongest statements are usually vertex-guard and vertex-count statements. They do not settle point-guard perimeter-over-six.

8.3 Perfect Graphs, Rectangles, and Perimeter

Maßberg's perimeter theorem sits in the rectangle/perfect-graph branch of the literature rather than the triangulation/coloring branch. The proof relates guarding to systems of maximal rectangles, uses perfect-graph structure to control the relevant packing/covering quantity, and then charges that structure to the lattice perimeter [19, Section 4]. This is why the theorem has the form $\max\{1, \lfloor \ell/6 \rfloor\}$ and why it is genuinely different from the $\lfloor n/4 \rfloor$ orthogonal-polygon theorem.

The same roadmap also explains the open hole case. In a rectilinear gallery with holes, the intersection graph of maximal rectangles need not retain the same perfect-graph behavior. Maßberg's habilitation identifies a maximal-rectangle packing conjecture as the missing structural substitute: for rectilinear galleries with holes, the maximum packing size of maximal rectangles should upper-bound the number of guards required [20, Conj. 6.10]. A proof of that conjecture, combined with the perimeter charging lemma for polyominoes with possible holes, would prove the perimeter theorem with holes [20, Lem. 6.9]; a counterexample would likely force either a correction term or a different invariant.

8.4 Cell-Count Polyomino Bounds

Biedl et al. prove the tight $\lfloor (m+1)/3 \rfloor$ theorem for connected m-cell polyominoes with $m \geq 2$, including holes [4, Cor. 1]. The lower bound is given by a comb family, and the upper bound follows from a constructive decomposition that also works under restrictive r-visibility [4, Fig. 3 and Theorem 1]. The one-cell case is handled separately by the obvious one guard. This proof line uses the cell structure directly, which is why it survives holes cleanly. The result is strong, but it answers a different extremal question: how guard number scales with area, not with boundary complexity. For research purposes, the important lesson is that holes are not inherently fatal in polyomino models; they are fatal to the specific perimeter proof technology currently known.

8.5 3D Decomposition and Guard Models

In three dimensions, the first modeling decision is guard type. The Paterson–Yao/Viglietta point-guard line shows that orthogonal polyhedra can require $\Theta(n^{3/2})$ point guards [24, 30]. That fact rules out a direct planar-style point-guard theorem in terms of vertices or edges.

The promising 3D analogy therefore uses edges. Benbernou et al. prove open-edge-guard bounds for orthogonal polyhedra, including bounds expressed in terms of total edges and reflex-edge count [3, Theorem 5 and Corollaries 6–7]; Viglietta proves tight-style results for 2-reflex orthogonal polyhedra [32]; and Cano–Toth–Urrutia–Viglietta improve the general polyhedron edge-guard bound [7]. The research gap is not whether edge guards are relevant, but whether the existing decomposition and charging methods can be sharpened to the lower-bound scale.

9 Status by Model

Simple orthogonal polygons, point guards.
Proven tight: every n-vertex simple orthogonal polygon is guardable by $\lfloor n/4 \rfloor$ point guards [16, 14].

Orthogonal polygons with holes, vertex guards.
Several upper bounds and partial sharp results are known. O'Rourke gives the classical $\lfloor (n+2h)/4 \rfloor$ style bound for h holes [23]. Shermer's stronger $\lfloor (n+h)/4 \rfloor$ conjecture is open in general; Zylinski gives a coloring proof of Aggarwal's theorem for $h \le 2$ and cactus-dual quadrilateralizations [29, 34]. In the h-independent direction, Hoffmann and Kriegel prove an $n/3$ upper bound, and Michael–Pinciu improve this line to $\lfloor (17n-8)/52 \rfloor$ using diagonal graphs and vertex covers [15, 21].

Hole-free polyominoes, point guards, lattice perimeter ℓ.
Proven tight: for $\ell \ge 6$, Maßberg proves that $\lfloor \ell/6 \rfloor$ point guards suffice [19, Theorem 1]. The exact theorem is $\max\{1, \lfloor \ell/6 \rfloor\}$, which includes the one-square case.

Polyominoes with holes, point guards, lattice perimeter ℓ.
Open/conditional. Maßberg's text explicitly states that the extension to holes would follow from a maximal-rectangle packing conjecture [20, Lem. 6.9 and Conj. 6.10].

Polyominoes with holes, point guards, area m.
Proven tight: every connected m-cell polyomino with $m \ge 2$, holes allowed, is guardable by $\lfloor (m+1)/3 \rfloor$ point guards [4, Cor. 1]. The one-cell case needs one guard. This does not settle perimeter-over-six.

Ortho-unit polygons, point guards.
Proven tight: every ortho-unit polygon with $n \ge 12$ vertices is guardable by $\lfloor (n-4)/8 \rfloor$ guards [9]. Since $n = P$ in the ortho-unit model, this is stronger than $P/6$ for that narrow class.

Integral orthogonal polygons, point guards, perimeter N.
Hole-free case proven via Maßberg: after subdividing long boundary edges into unit grid edges, a hole-free integral orthogonal polygon is a hole-free polyomino with $N = \ell$. Diaz-Banez et al. state the natural $N/6$ conjecture in the broader integral-perimeter discussion [9]; its unresolved content is therefore beyond the hole-free case, notably holes.

Polycubes and polyhypercubes.
Pinciu proves the higher-dimensional cell-count analogue: every m-polyhypercube in dimension $d \ge 2$ is guardable by $\lfloor (m+1)/3 \rfloor$ point guards, and some examples require that many [25]. For $d = 3$ this is the polycube version of the polyomino area theorem. It is still a unit-cell theorem, not a theorem for arbitrary continuous orthogonal polyhedra.

Orthogonal polyhedra, point guards.
No planar-style linear theorem exists in terms of vertices. The Paterson–Yao bound gives $\Theta(n^{3/2})$ point guards as sufficient and sometimes necessary for n-vertex orthogonal polyhedra [24, 30].

Orthogonal polyhedra, edge and reflex-edge guards.
The central open direction is an Urrutia-type edge-guard theorem: lower-bound examples require roughly $m/12$ edge guards, while known upper bounds are larger. Benbernou et al. prove that $\lfloor (m+r)/12 \rfloor$ open edge guards suffice for an orthogonal polyhedron with m edges and r reflex edges; their corollaries give $(11/72)m - g/6 - 1$ and $(7/12)r - g + 1$ open-edge-guard bounds [3, Theorem 5 and Corollaries 6–7].

The latter is a bound parameterized by reflex-edge count, not a reflex-edge-only guard theorem. Viglietta proves optimal-style bounds for 2-reflex orthogonal polyhedra [32]. Cano, Toth, Urrutia, and Viglietta improve the general 3D polyhedron edge-guard bound to $(5/6)m$, while noting a substantial remaining gap [7]. Aldana-Galvan et al. prove a related $\pi/2$-edge-guard theorem for orthogonal polyhedra [2]; this is adjacent limited-field context, not a standard edge-guard or reflex-edge-guard closure.

10 Two-Dimensional Orthogonal Polygons

10.1 Classical Background

The original art-gallery theorem is due to Chvatal: $\lfloor n/3 \rfloor$ guards always suffice for a simple n-vertex polygon, and this is tight [8]. Fisk's short proof via triangulation and three-coloring became the standard proof template [12]. O'Rourke's book remains the standard comprehensive reference for the classical theory [23]; Shermer and Urrutia survey art-gallery and illumination problems, including variants and open problems [27, 28]. On the algorithmic side, Abrahamsen, Adamaszek, and Miltzow prove that the general art-gallery problem is $\exists\mathbb{R}$-complete even for simple polygons with integer-coordinate vertices, underscoring why restricted orthogonal and lattice models are usually treated separately [1, Theorem 1].

10.2 Simple Orthogonal Polygons

Kahn, Klawe, and Kleitman prove the tight orthogonal theorem: $\lfloor n/4 \rfloor$ point guards suffice for every simple orthogonal polygon with n vertices [16]. Gyori gives a short proof [14]. This theorem is a vertex-count theorem, not a perimeter theorem. In a lattice setting the vertex count and the perimeter may be very different, which is why the polyomino and integral-perimeter questions are separate.

The proof tradition connects to decomposition, coloring, and perfect-graph methods. Motwani, Raghunathan, and Saran study covering orthogonal polygons with star polygons [22]; Worman and Keil develop decomposition and visibility-graph machinery for orthogonal art-gallery problems [33]. Bjørling-Sachs proves the tight $\lfloor n/4 \rfloor$ edge-guard bound for rectilinear polygons, a useful 2D reference point for later edge-guard models [6].

10.3 Holes

Holes are one of the main fault lines in the literature. For orthogonal polygons with h holes, classical vertex-guard bounds exist, but the strongest natural conjectures are not settled in general. Urrutia's open-problem list records Shermer's conjecture that an n-vertex orthogonal polygon with h holes should be guardable by $\lfloor (n+h)/4 \rfloor$ vertex guards, and Hoffmann's conjecture that $\lfloor 2n/7 \rfloor$ vertex guards should suffice for orthogonal polygons with holes [29]. Zylinski proves the $\lfloor (n+h)/4 \rfloor$ bound under a cactus-dual quadrilateralization condition and recovers Aggarwal's theorem for $h \le 2$ [34]. Hoffmann and Kriegel prove graph-coloring consequences for rectilinear polygons with holes, including upper bounds for rectilinear prison-yard variants [15]. Michael and Pinciu later reframe much of this orthogonal vertex-guard theory through same-sign diagonal graphs and vertex covers; in particular, they give an h-independent upper bound $\lfloor (17n-8)/52 \rfloor$ for orthogonal polygons with any number of holes [21].

These hole results are relevant context, but they do not prove the perimeter-over-six statement. They usually concern vertex guards and vertex counts; the perimeter-over-six problem concerns point guards and lattice perimeter.

11 Polyomino and Perimeter Results

11.1 Hole-Free Polyominoes

Maßberg proves the direct perimeter theorem for hole-free polyominoes [19, Theorem 1]. In the notation of that paper, if a hole-free polyomino has lattice perimeter ℓ, then it can be guarded by at most $\max\{1, \lfloor \ell/6 \rfloor\}$

point guards; for $\ell \geq 6$ this is exactly $\lfloor \ell/6 \rfloor$. The proof uses maximal axis-parallel rectangles in a rectilinear gallery, relates a rectangle packing number to the guard number, and then bounds the packing by the perimeter. A comb construction shows tightness [19, Fig. 3 and p. 575].

11.2 Polyominoes with Holes

The area theorem does allow holes: Biedl, Irfan, Iwerks, Kim, and Mitchell prove that every connected m-cell polyomino with $m \geq 2$ can be guarded by $\lfloor (m+1)/3 \rfloor$ point guards, and the bound is tight [4, Cor. 1]. The one-cell case is trivial, and the sufficiency proof works even under restrictive r-visibility [4, Theorem 1]. This is the strongest clean theorem for arbitrary polyominoes with holes in the standard point-guard model, but it is an area bound rather than a perimeter bound.

For perimeter, the hole case remains conditional. Maßberg observes that the perimeter theorem would extend to holes if a maximal-rectangle packing conjecture were true [20, Conj. 6.10]. The thesis also proves the perimeter charging lemma for maximum packings in polyominoes with possible holes [20, Lem. 6.9]. This is *not* a proof of perimeter-over-six for holes; it identifies the missing structural statement.

11.3 Ortho-Unit and Integral Orthogonal Polygons

Diaz-Banez et al. prove that every ortho-unit polygon with $n \geq 12$ vertices can be guarded by $\lfloor (n-4)/8 \rfloor$ guards, and that the bound is tight [9]. Since every edge has unit length in an ortho-unit polygon, n is the perimeter. This solves a narrow unit-edge class more strongly than $P/6$.

The same paper frames the integral-perimeter conjecture: every integral orthogonal polygon of perimeter N should be guardable by $\lfloor N/6 \rfloor$ point guards. If "integral orthogonal polygon" is read in the usual simple, hole-free sense, this is already covered by Maßberg. Subdividing long sides into unit grid edges turns the polygon into a hole-free polyomino with the same lattice perimeter, $N = \ell$, and Maßberg proves $\max\{1, \lfloor \ell/6 \rfloor\}$ for that class [19, Theorem 1]. Thus the Diaz-Banez question is open only under a broader reading, such as integral domains with holes or another lattice-domain convention not reduced to a hole-free polyomino.

12 Three-Dimensional Orthogonal Polyhedra

12.1 Point Guards

The direct 3D point-guard analogue of planar art-gallery theorems fails. In orthogonal polyhedra with n vertices, Paterson and Yao's binary-space partition result implies a tight $\Theta(n^{3/2})$ point-guard bound [24]. Viglietta's thesis presents this as Theorem 3.7 and draws the key modeling conclusion: point guards do not yield a linear orthogonal-polyhedron art-gallery theorem [30]. Therefore a 3D "perimeter-over-six" theorem is not obtained by simply replacing polygons with orthogonal polyhedra and keeping point guards.

12.2 Edge and Reflex-Edge Guards

The 3D literature often treats edge guards as the analogue of planar vertex guards. Urrutia conjectured that an orthogonal polyhedron with m edges should be guardable by $m/12 + O(1)$ closed edge guards, matching known lower bound examples up to an additive constant [30]. A related reflex-edge conjecture asks whether roughly $(r-g)/2 + 1$ reflex edge guards suffice, where r is the number of reflex edges and g is the genus.

Benbernou, Demaine, Demaine, Kurdia, O'Rourke, Toussaint, Urrutia, and Viglietta improve the known asymptotic upper bounds for orthogonal polyhedra: their main open-edge theorem gives $\lfloor (m+r)/12 \rfloor$ guards, and their corollaries yield $(11/72)m - g/6 - 1$ in terms of total edges and $(7/12)r - g + 1$ in terms of reflex-edge count [3, Theorem 5 and Corollaries 6–7]. This is an edge-guard theorem parameterized by r, not a reflex-edge-only guard theorem. Viglietta later proves optimal-style bounds for 2-reflex orthogonal polyhedra, namely orthogonal polyhedra whose reflex edges occur in only two directions [32]. These are major partial results, but they do not settle the general 3-reflex orthogonal-polyhedron conjectures.

Cano, Toth, Urrutia, and Viglietta give a broad theorem for arbitrary polyhedra in three-space: every polyhedron with m edges can be guarded by at most $(5/6)m$ edge guards [7]. This improves the general-polyhedron edge-guard bound, but it still leaves a large gap from the conjectured $m/6$ style lower/upper behavior for general polyhedra and the $m/12 + O(1)$ orthogonal-polyhedron conjecture.

Aldana-Galvan, Alvarez-Rebollar, Catana-Salazar, Jimenez-Salinas, Solis-Villarreal, and Urrutia prove a theorem for $\pi/2$-edge guards in orthogonal polyhedra [2]. This source belongs in the 3D edge-guard neighborhood, but the guard has limited field of view, so it is recorded as a separate adjacent model rather than evidence for the standard closed-edge or reflex-edge conjectures.

12.3 Face Guards and Polycube Variants

Face guards are another 3D model. Viglietta studies face-guarding for several classes of polyhedra, including orthogonal polyhedra, 4-oriented polyhedra, and 2-reflex orthostacks [31]. Face guards are useful for mapping the 3D model space, but they are not a direct continuation of the point-guard perimeter-over-six question.

For lattice 3D objects, Pinciu's theorem gives the direct point-guard cell-volume analogue: $\lfloor (m+1)/3 \rfloor$ point guards are tight for m-polyhypercubes in every dimension $d \geq 2$ [25]. Pinciu's separate pixel/voxel-guard bounds remain a discrete cell-guard line with an explicit sharpness conjecture, so the paper keeps point guards and cell guards in separate model cells.

13 Algorithmic and Optimization Context

Extremal art-gallery theorems answer a worst-case existence question: how many guards always suffice for every object in a class? Optimization papers ask a different question: given one object, how hard is it to find a minimum guard set? The distinction matters because a clean universal upper bound need not yield an efficient exact algorithm for the minimum set, and an NP-hardness result does not rule out a sharp extremal theorem.

Lee and Lin established the classical NP-hardness framework for art-gallery optimization problems [18]. In orthogonal settings, Schuchardt–Hecker prove NP-hardness for two orthogonal art-gallery problems [26], and Katz–Roisman study hardness for guarding vertices of rectilinear domains [17]. At the level of the general point-guard problem, Abrahamsen–Adamaszek–Miltzow show $\exists\mathbb{R}$-completeness, already for simple polygons with integer-coordinate vertices [1, Theorem 1]. These complexity results should be read as optimization barriers, not as barriers to universal counting theorems such as $\lfloor n/4 \rfloor$ or $\lfloor \ell/6 \rfloor$.

Approximation and restricted-visibility models form a parallel literature. Ghosh gives broad approximation-algorithm context for art-gallery problems [13]. Durocher et al. and Biedl et al. study sliding cameras and sliding k-transmitters in orthogonal polygons, including holes and hardness or approximation results in those models [10, 5]. Filtser et al. study polyominoes under k-hop visibility, a discrete variant close to network or city-block applications rather than straight-line point-guard visibility [11]. These variants are useful sources of techniques and open problems, but they should not be cited as settling the standard perimeter-over-six question.

14 Open Problems and Research Directions

The following problems are the most directly connected to the survey above. Items are grouped by mathematical model rather than by perceived difficulty. When a question is only a consequence of comparing known results, the text says so explicitly; otherwise the cited source records the conjecture or the conditional dependency.

Two entries below are different in kind. They are recorded as working prompts for Galen from the current project, not as literature-confirmed open problems: Paul's flat-vertex hybrid expression and a possible 3D surface-area-over-eight analogue. They are included because they are useful research directions, but the table and prose mark the missing model choices explicitly.

1. **Perimeter-over-six for polyominoes with holes.** Let P be a polyomino with holes and total lattice perimeter ℓ, counting both outer and hole boundaries. Is P always guardable by $\max\{1, \lfloor \ell/6 \rfloor\}$ point

Problem cluster	Status source	What is known	What would constitute progress
Polyomino perimeter with holes	Maßberg habilitation [20, Conj. 6.10]	Hole-free theorem is published; hole extension is conditional on a maximal-rectangle packing conjecture.	Prove the packing conjecture, prove $\lfloor \ell/6 \rfloor$ by another method, or construct a hole counterexample.
Integral perimeter beyond the hole-free case	Diaz-Banez et al.; Maßberg [9, 19]	Hole-free integral polygons reduce to hole-free polyominoes.	State and resolve the exact holes/broader-domain version, including how perimeter is counted.
Paul/Galen flat-vertex hybrid	Project working question, not source-stated	If $P = n + f$, then $n/8 + f/4 < P/6$ is equivalent to $f < n/2$; no source was found for a guard theorem using this hybrid parameter.	Fix f, holes, and guard model; decide whether the target is an upper bound, lower-bound family, or improvement regime.
Orthogonal polygons with holes, vertex guards	Urrutia; Zylinski; Michael–Pinciu [29, 34, 21]	Strong conjectures remain open; partial and h-independent upper bounds are known.	Close the gap between lower-bound constructions and the best vertex-guard upper bounds.
3D orthogonal edge guards	Viglietta; Benbernou et al. [30, 3]	Known upper bounds exceed Urrutia-type lower-bound targets.	Prove an $m/12 + O(1)$ style theorem or identify a larger necessary correction term.
3D reflex-edge guards	Viglietta [32]	2-reflex orthogonal polyhedra are understood much better than general 3-reflex orthogonal polyhedra.	Extend the induction/charging methods beyond the 2-reflex restriction.
3D surface-area-over-eight	Project working conjecture, not source-stated	The question needs lattice-normalized surface area; face-count and volume-count theorems are separate model cells.	Choose object class and guard model, then test lower- and upper-bound constructions.
Polycube/polyhypercube voxel guards	Pinciu [25]	Point-guard cell-volume bounds are tight, but pixel/voxel guard bounds have only partial formulas and a sharpness conjecture.	Prove or refute the dependent-on-d pixel/voxel-guard conjecture.
Algorithmic restricted cases	Lee–Lin; Schuchardt–Hecker; Ghosh [18, 26, 13]	Exact optimization is hard in many settings; approximation and special visibility models remain active.	Find tractable subclasses aligned with the extremal perimeter or edge-guard conjectures.

Table 3: *Research-facing status table for the main open-problem clusters.*

guards? Maßberg's habilitation gives a conditional route to this theorem through a maximal-rectangle packing conjecture [20, Lem. 6.9 and Conj. 6.10]; the published perimeter theorem is the hole-free case [19, Theorem 1].

2. **Maßberg's maximal-rectangle packing conjecture.** In a rectilinear gallery with holes, does the maximum packing size of maximal rectangles control the required number of point guards? This is the structural conjecture identified by Maßberg as sufficient for extending the perimeter-over-six theorem to polyominoes with holes [20, Conj. 6.10]. The same discussion notes that maximal-rectangle intersection graphs need

not be perfect in the hole case, so the conjecture is not a routine extension of the hole-free perfect-graph argument.

3. **Integral-perimeter bounds beyond the hole-free case.** The hole-free integral orthogonal polygon version follows from Maßberg by unit-grid subdivision. The remaining integral-perimeter question is whether the same $\lfloor N/6 \rfloor$ behavior survives when holes or broader lattice-domain conventions are allowed [19, 9].

4. **Paul/Galen flat-vertex hybrid question.** This is a project working prompt rather than a source-stated open problem. Suppose an integral orthogonal polygon has been subdivided into unit boundary steps, with total lattice perimeter P, n turn vertices, and f flat subdivision vertices. Under the convention $P = n + f$,

$$n/8 + f/4 < P/6 \iff f < n/2.$$

Thus the inequality is not itself the mathematical content. The real question is whether the hybrid expression $n/8 + f/4$ controls a guard theorem, lower-bound family, or useful improvement regime once f, holes, and guard model are fixed. The current evidence pass found no primary source for an $n/8 + f/4$ art-gallery theorem or source-stated open problem.

5. **Vertex guards for orthogonal polygons with holes.** Shermer's conjecture asks whether every n-vertex orthogonal polygon with h holes is guardable by $\lfloor (n + h)/4 \rfloor$ vertex guards. Hoffmann's conjecture asks whether $\lfloor 2n/7 \rfloor$ vertex guards always suffice. These are source-stated open problems in the orthogonal-with-holes literature, and they are related to but distinct from perimeter-type point-guard bounds [29, 15, 34, 21].

6. **A correction term if perimeter-over-six fails with holes.** This is a derived direction rather than a named conjecture. If holes force counterexamples to a pure $\ell/6$ theorem, a useful replacement may need to account for total perimeter, outer perimeter, number of holes, hole perimeter, reflex vertices, or rectangle-packing obstructions.

7. **Urrutia-type edge-guard bounds for orthogonal polyhedra.** For genus-zero orthogonal polyhedra with m edges, the central conjectural target is an $m/12 + O(1)$ edge-guard bound. Lower-bound examples and current upper bounds leave a substantial gap [30, 3].

8. **Reflex-edge guards in general orthogonal polyhedra.** Viglietta proves tight-style bounds for 2-reflex orthogonal polyhedra, where reflex edges occur in only two coordinate directions, but the analogous reflex-edge conjecture for general orthogonal polyhedra remains open [32].

9. **The general three-dimensional edge-guard gap.** Cano, Toth, Urrutia, and Viglietta improve the edge-guard upper bound for arbitrary polyhedra to $(5/6)m$, but explicitly leave the gap between known lower and upper bounds as an open problem [7]. Orthogonal polyhedra form the more structured subclass where sharper Urrutia-type conjectures are expected.

10. **Surface-area-over-eight in 3D.** This is also a project working conjecture rather than a literature-confirmed theorem. A meaningful statement must first choose a lattice-normalized surface area S, such as the number of exposed unit square faces of a polycube or grid-refined orthogonal polyhedron, and must specify whether the guards are points, vertices, edges, faces, or unit-face/cell guards. Nearby sources do not settle this: Viglietta's face-guard paper uses a different guard model [31], Pinciu's polyhypercube theorem is a volume/cell-count point-guard result [25], and continuous 3D point guarding has different vertex-count behavior [24, 30].

11. **Pixel/voxel guards for polyhypercubes.** Pinciu settles the point-guard cell-volume question for every dimension $d \geq 2$, but the corresponding pixel/voxel-guard model is different. In that model Pinciu gives

lower bounds, dimension-independent upper bounds, and a conjectured sharp dependent-on-d formula [25]. This is an open problem for discrete cell guards, not a gap in the point-guard $\lfloor (m+1)/3 \rfloor$ theorem.

12. **Algorithmic complexity and approximation in restricted models.** Extremal guard bounds and optimization algorithms remain separate questions. Minimum guarding is NP-hard in many orthogonal settings [18, 26, 17, 4, 10, 5]. A parallel algorithmic direction is to identify exact polynomial subclasses and approximation guarantees for polyominoes with holes, integral orthogonal domains, polycubes, and orthogonal polyhedra. Recent work on k-hop visibility in polyominoes records related approximation questions [11].

15 Annotated Source Guide

Foundational Art-Gallery Theory

Chvatal [8] and Fisk [12].
Foundational $\lfloor n/3 \rfloor$ theorem for simple polygons and the short triangulation/coloring proof.

O'Rourke [23].
Standard book reference for art-gallery theorems, algorithms, visibility, decompositions, and early 3D context.

Shermer [27] and Urrutia [28].
Survey references for art-gallery and illumination problems, useful for terminology, variants, and classical open-problem context.

Lee–Lin [18].
Classical computational-complexity source for art-gallery optimization problems.

Abrahamsen–Adamaszek–Miltzow [1].
Modern complexity reference proving $\exists\mathbb{R}$-completeness for the general art-gallery problem, even for simple integer-coordinate polygons.

2D Orthogonal Polygons and Holes

Kahn–Klawe–Kleitman [16].
Primary source for the tight $\lfloor n/4 \rfloor$ theorem for simple orthogonal polygons.

Gyori [14].
Short proof of the rectilinear art-gallery theorem.

Hoffmann–Kriegel [15].
Graph-coloring result with consequences for rectilinear polygons with holes and prison-yard variants.

Zylinski [34].
Coloring proof of Aggarwal's theorem for orthogonal galleries with holes, including the $\lfloor (n+h)/4 \rfloor$ bound for $h \le 2$ and cactus-dual cases.

Michael–Pinciu [21].
Same-sign diagonal graph and vertex-cover framework for orthogonal polygons, including h-independent bounds for polygons with holes.

Urrutia open-problem page [29].
Direct source for Shermer's and Hoffmann's open conjectures on orthogonal polygons with holes using vertex guards.

Schuchardt–Hecker [26].
NP-hardness source for art-gallery problems in orthogonal polygons.

Worman–Keil [33].
Decomposition and visibility-graph machinery for the orthogonal art-gallery problem.

Bjørling-Sachs [6].
Tight edge-guard theorem for rectilinear polygons; relevant when comparing 2D orthogonal edge guards with 3D edge-guard conjectures.

Katz–Roisman [17].
NP-hardness for guarding vertices of rectilinear domains by vertex, boundary, or point guards.

Polyomino, Perimeter, and Lattice Variants

Maßberg [19].
Primary peer-reviewed source for perimeter-over-six in hole-free polyominoes: Theorem 1 gives $\max\{1, \lfloor \ell/6 \rfloor\}$ and Fig. 3 gives the tight comb construction. This also proves the nontrivial hole-free integral orthogonal polygon $\lfloor N/6 \rfloor$ statement after unit-grid subdivision.

Maßberg habilitation [20].
Source for the conditional hole extension. Lemma 6.9 gives the perimeter charging statement for maximum rectangle packings in polyominoes with possible holes, and Conjecture 6.10 is the missing guard-number statement for rectilinear galleries with holes.

Biedl–Irfan–Iwerks–Kim–Mitchell [4].
Tight $\lfloor (m+1)/3 \rfloor$ cell-count theorem for connected m-polyominoes with $m \geq 2$, holes allowed. The definitions are on p. 712, the comb lower bound is Fig. 3, and Corollary 1 gives the theorem.

Diaz-Banez et al. [9].
Recent tight theorem for ortho-unit polygons and source of the integral orthogonal polygon $\lfloor N/6 \rfloor$ conjecture. Its simple/hole-free reading overlaps with Maßberg rather than creating a separate open problem.

Pinciu [25].
Point-guard cell-count theorem: for every $d \geq 2$, $\lfloor (m+1)/3 \rfloor$ guards are tight for m-polyhypercubes. The same source also records separate pixel/voxel-guard bounds and a sharpness conjecture.

Filtser–Krohn–Nilsson–Rieck–Schmidt [11].
Current work on k-hop visibility in polyominoes, including open approximation questions for polyominoes with holes.

3D Orthogonal Polyhedra

Paterson–Yao [24].
Binary-space partition result used for the $\Theta(n^{3/2})$ point-guard bound in orthogonal polyhedra.

Viglietta thesis [30].
Central survey and source for guarding/searching polyhedra, the point-guard obstruction, and Urrutia-type edge-guard conjectures.

Benbernou et al. [3].
CCCG source for improved edge-guard bounds in orthogonal polyhedra: $\lfloor (m+r)/12 \rfloor$ open edge guards,

with corollaries $(11/72)m - g/6 - 1$ and $(7/12)r - g + 1$. The r bound is parameterized by reflex-edge count; it is not a reflex-edge-only guard theorem.

Aldana-Galvan et al. [2].
Adjacent CCCG source for $\pi/2$-edge guards in orthogonal polyhedra. It is useful limited-field context, but it should not be read as a standard edge-guard or reflex-edge-guard theorem.

Viglietta 2-reflex paper [32].
Proves optimal-style reflex-edge bounds for 2-reflex orthogonal polyhedra and gives an $O(n \log n)$ guard-placement algorithm.

Viglietta face-guarding paper [31].
Face-guarding results for several classes of polyhedra, including orthogonal subclasses.

Cano–Toth–Urrutia–Viglietta [7].
General edge-guard result in three-space and explicit open gap.

Algorithms and Related Visibility Models

Motwani–Raghunathan–Saran [22].
Perfect-graph approach to covering orthogonal polygons with star polygons.

Ghosh [13].
General approximation-algorithm context for art-gallery problems.

Durocher et al. [10].
Sliding-camera model for orthogonal art galleries, including holes.

Biedl et al. [5].
Sliding k-transmitter hardness and approximation results for orthogonal polygons.

16 Completeness Protocol for This Review

This review treats completeness as an auditable process rather than as a claim that any finite search has exhausted the literature. A result should enter the narrative only after its source, exact model, theorem or problem statement, safe translation, coverage status, and open-problem consequences have been recorded. This discipline matters here because small changes in holes, lattice assumptions, guard type, dimension, and parameter frequently turn a valid theorem into a non-comparable statement.

Step	Question	Review artifact
1	What is the exact scope cell? Record object class, holes, guard model, dimension, and parameter before comparing results.	Coverage matrix row or new coverage cell.
2	What are the seed sources? Start from books, surveys, theses, open-problem pages, and named theorem papers.	Source-registry records with primary DOI, publisher, arXiv, or author links when available.
3	What searches have been run? Use exact-title searches, DOI/arXiv lookups, author pages, backward references, forward citation searches, venue scans, and terminology variants.	Search-log entries with included sources, exclusions, and follow-up actions.
4	Is the source primary for the claim? Prefer the paper that proves or states the result; use surveys and books to orient the search and explain history.	Source type and role recorded in the source registry.
5	What exactly is the claim? Extract theorem, conjecture, lower bound, hardness, algorithmic, survey, or open-problem records without changing the model.	Claim-registry entry with status, model, parameter, locator, and citation.
6	What does the claim not imply? Record blocked translations across holes, polyominoes, integral polygons, guard models, and 3D analogues.	Translation note in the claim registry and, when central, prose in the model-separation section.
7	Which status changed? Update every affected coverage cell as proved, tight, conditional, open, synthesized, out of scope, or still needing follow-up.	Coverage matrix with supporting claim identifiers and next action.
8	Is an open problem source-stated, conditional, or derived? Keep these categories separate and record dependencies.	Open-problem ledger linked to supporting sources and claims.
9	Are locators and quotations sufficient? Add theorem numbers, page numbers, figure references, or short quotations for problem statements where available.	Evidence-report gap list should shrink after each locator pass.
10	Does the document still assemble cleanly? Check that the short answer, status maps, proof discussion, open problems, source guide, and evidence trail still form one reader-facing flow.	Document assembly map and paper outline.
11	Is the framework progress current? Record which stage changed, what remains incomplete, and which artifact should be updated next.	Progress tracker and current work queue.
12	Has a final status pass been run for central open, conditional, or recent-status claims? Repeat forward-citation, venue, author-page, exact-title, and adjacent-model searches before preserving a status.	Search-log and search-query entries with dated outcomes for each central status cell.
13	Is the prose synchronized with the evidence? Regenerate the evidence report, run registry checks, rebuild the PDF, and inspect the log before finalizing.	Updated paper, generated evidence report, and clean validation run.

Completeness Gates

The practical standard for this paper is not "no missing papers." It is that every central status statement can be audited. In particular, a claim that a problem is open should be supported by one of three paths: an explicit open-problem statement in a primary or survey source, a conditional theorem whose stated hypothesis remains unproved in the tracked sources, or a derived gap visible from the coverage matrix after the relevant model cells have been searched. The last category should be phrased cautiously and should identify the completed searches that make the gap credible.

The current framework therefore makes gaps visible instead of hiding them in prose. At the time of this pass, every claim record has a locator at the level currently available from primary PDFs, open-problem pages, public technical reports, or primary metadata. The forward/adjacent sweep and final OpenAlex/web pass for the central holes, perimeter, and 3D edge/reflex-edge clusters are now logged. They did not add a tracked theorem that changes the main statuses. It also records two project working questions without promoting them to literature-confirmed open problems: Paul's $n/8 + f/4 < P/6$ flat-vertex prompt and a 3D surface-area-over-eight prompt. The current coverage matrix now separates the cited adjacent variants that already have source records: two-dimensional edge guards, sliding-camera and sliding-transmitter visibility, k-hop visibility, polyhypercube point guards, polyhypercube pixel/voxel guards, 3D face guards, $\pi/2$-edge guards, and 2-reflex reflex-edge guards. It also records scope-exclusion cells for floodlight, half-plane, dispersive, contiguous, mobile, point-boundary, and related variants. Those cells are not source claims; they are checkpoints requiring source intake before any such variant is used as a theorem-level result. The remaining completeness work is recurring search-frontier work and model-confirmation work rather than a missing-locator problem.

17 Evidence Audit Trail

The review is backed by a structured evidence layer in the repository. The source registry records bibliographic information and links; the claims registry records the specific theorem, conjecture, hardness, survey, or open-problem statement extracted from each source; the coverage matrix records which model cells are proved, open, conditional, synthesized, or still in need of follow-up; and the open-problems ledger records the unresolved questions and their supporting sources.

The generated evidence report is intended as an audit companion to the prose. It lists central proved results, open and conditional problems, coverage cells, missing page or theorem locators, and next actions for improving the review. The document assembly map records how the evidence artifacts become a readable paper, and the framework progress tracker records which part of the review process is current, which part needs work, and which artifact should be touched next. The prose above should be updated only after the corresponding source, claim, coverage, open-problem, assembly, and progress records support the statement being added.

The May 23, 2026 search-log entries are the current checkpoint for recent-status claims. They record exact-title, OpenAlex forward-citation, arXiv/DROPS/topic, venue, and adjacent-model queries for the orthogonal-polygons-with-holes, polyomino-perimeter-with-holes, integral perimeter, and 3D orthogonal-polyhedron clusters. The result was mainly status preservation: the central open and conditional cells remain open or conditional in the tracked evidence. The one added theorem source is Aldana-Galvan et al.'s $\pi/2$-edge-guard result, which is recorded as a separate adjacent model rather than as a closure of the standard 3D edge/reflex-edge gaps.

The May 26, 2026 Galen/Paul working-question pass records two proposed questions separately from source-stated open problems. No source was found for an $n/8 + f/4$ guard theorem, an $n/8 + f/4 < P/6$ open problem, or a surface-area-over-eight theorem for 3D orthogonal objects. The evidence layer therefore treats both prompts as model-definition tasks first: fix the flat-vertex parameter and intended guard statement in 2D, and fix the surface normalization, object class, and guard model in 3D.

Cell checked	Outcome	Reader-facing consequence
Polyominoes with holes, point guards, lattice perimeter ℓ	Status preserved	No tracked source proves the $\ell/6$ holes extension; it remains tied to Maßberg's conditional route.
Integral orthogonal polygons, point guards, perimeter N	Status preserved	The hole-free reading remains covered by Maßberg; broader readings with holes remain open or conditional in the tracked evidence.
Orthogonal polygons with holes, vertex guards	Status preserved	Floodlight and half-plane hits remain adjacent exclusions; no standard vertex-guard resolution was tracked.
Orthogonal polyhedra, standard edge/reflex-edge guards	Adjacent source added; status preserved	Aldana-Galvan et al. add $\pi/2$-edge-guard context, but the standard edge/reflex-edge gaps remain open.
Paul/Galen flat-vertex and surface-area prompts	Working questions recorded; no source-status change	The prompts are useful research directions, but the PDF treats them as model-definition tasks until the missing conventions or sources are supplied.

18 Conclusion

1. The theorem "$\ell/6$ guards suffice" is established for *hole-free polyominoes*. The polyomino-with-holes version is conditional on Maßberg's rectangle-packing conjecture in the sources surveyed.

2. The statement "an integral orthogonal polygon of perimeter N is guardable by $\lfloor N/6 \rfloor$ guards" has two different readings. The simple/hole-free reading is already proved by Maßberg, because it is exactly the hole-free polyomino case after unit-grid subdivision. The holes or broader-domain reading remains open/conditional.

3. The right 3D terminology is *orthogonal polyhedron* or *orthogonal polytope*. The natural 3D guard models are not obtained by copying the 2D point-guard setting.

4. A 3D point-guard analogue of the planar perimeter theorem is blocked by the $\Theta(n^{3/2})$ point-guard bound. The central 3D problems instead concern edge guards, reflex-edge guards, face guards, or discrete polycube visibility models.

5. The most direct open directions are the polyomino-with-holes perimeter bound, Maßberg's rectangle-packing conjecture, the holes/broader-domain $\lfloor N/6 \rfloor$ extension, and the Urrutia-type $m/12 + O(1)$ and reflex-edge conjectures for general orthogonal polyhedra.

6. Paul's $n/8 + f/4 < P/6$ prompt and the 3D surface-area-over-eight prompt are now recorded as proposed working questions for Galen. They should not be cited as literature-confirmed open problems until their model conventions or source status are fixed.

References

[1] M. Abrahamsen, A. Adamaszek, and T. Miltzow. The art gallery problem is $\exists\mathbb{R}$-complete. *Journal of the ACM*, 69(1), Article 4, 2022. https://doi.org/10.1145/3486220

[2] I. Aldana-Galvan, J. L. Alvarez-Rebollar, J. C. Catana-Salazar, M. Jimenez-Salinas, E. Solis-Villarreal, and J. Urrutia. Minimizing the solid angle sum of orthogonal polyhedra and guarding them with $\pi/2$-edge guards. In *Proceedings of the 28th Canadian Conference on Computational Geometry*, 2016. https://mazayjimenez.org/papers/minimizing_cccg2016.pdf https://www.cccg.ca/proceedings/2016/proceedings2016.pdf

[3] N. M. Benbernou, E. D. Demaine, M. L. Demaine, A. Kurdia, J. O'Rourke, G. T. Toussaint, J. Urrutia, and G. Viglietta. Edge-guarding orthogonal polyhedra. In *Proceedings of the 23rd Canadian Conference*

on Computational Geometry, 461–466, 2011. https://erikdemaine.org/papers/EdgeGuarding_CCCG2011/ https://www.science.smith.edu/~jorourke/Papers/EdgeGuarding.pdf

[4] T. C. Biedl, M. T. Irfan, J. Iwerks, J. Kim, and J. S. B. Mitchell. The art gallery theorem for polyominoes. *Discrete & Computational Geometry*, 48, 711–720, 2012. https://doi.org/10.1007/s00454-012-9429-1 https://tildesites.bowdoin.edu/~mirfan/papers/Art_Gallery_DCG_2012.pdf https://researchconnect.suny.edu/en/publications/the-art-gallery-theorem-for-polyominoes/

[5] T. Biedl, T. M. Chan, S. Lee, S. Mehrabi, F. Montecchiani, H. Vosoughpour, and Z. Yu. Guarding orthogonal art galleries with sliding k-transmitters: hardness and approximation. *Algorithmica*, 81(1), 69–97, 2019. https://doi.org/10.1007/s00453-018-0433-6

[6] I. Bjørling-Sachs. Edge guards in rectilinear polygons. *Computational Geometry: Theory and Applications*, 11(2), 111–123, 1998. https://doi.org/10.1016/S0925-7721(98)00024-8

[7] J. Cano, C. D. Toth, J. Urrutia, and G. Viglietta. Edge guards for polyhedra in three-space. *Computational Geometry: Theory and Applications*, 104, 101859, 2022. https://doi.org/10.1016/j.comgeo.2022.101859 https://giovanniviglietta.com/papers/edgeguards.pdf

[8] V. Chvatal. A combinatorial theorem in plane geometry. *Journal of Combinatorial Theory, Series B*, 18(1), 39–41, 1975. https://doi.org/10.1016/0095-8956(75)90061-1

[9] J. M. Diaz-Banez, P. Horn, M. A. Lopez, N. Marin, A. Ramirez-Vigueras, O. Sole-Pi, A. Stevens, and J. Urrutia. Ortho-unit polygons can be guarded with at most $\lfloor(n-4)/8\rfloor$ guards. *Graphs and Combinatorics*, 41, 15, 2025. https://doi.org/10.1007/s00373-024-02880-8 https://arxiv.org/abs/2208.12864

[10] S. Durocher, O. Filtser, R. Fraser, A. D. Mehrabi, and S. Mehrabi. Guarding orthogonal art galleries with sliding cameras. *Computational Geometry: Theory and Applications*, 65, 12–26, 2017. https://doi.org/10.1016/j.comgeo.2017.04.001

[11] O. Filtser, E. Krohn, B. J. Nilsson, C. Rieck, and C. Schmidt. Guarding polyominoes under k-hop visibility. *Algorithmica*, 87, 572–593, 2025. https://doi.org/10.1007/s00453-024-01292-7

[12] S. Fisk. A short proof of Chvatal's watchman theorem. *Journal of Combinatorial Theory, Series B*, 24(3), 374, 1978. https://doi.org/10.1016/0095-8956(78)90059-X

[13] S. K. Ghosh. Approximation algorithms for art gallery problems in polygons. *Discrete Applied Mathematics*, 158(6):718–722, 2010. https://doi.org/10.1016/j.dam.2009.12.004

[14] E. Gyori. A short proof of the rectilinear art gallery theorem. *SIAM Journal on Algebraic and Discrete Methods*, 7(3), 452–454, 1986. https://doi.org/10.1137/0607051

[15] F. Hoffmann and K. Kriegel. A graph-coloring result and its consequences for polygon-guarding problems. *SIAM Journal on Discrete Mathematics*, 9(2), 210–224, 1996. https://doi.org/10.1137/S0895480194265611 https://refubium.fu-berlin.de/handle/fub188/18382

[16] J. Kahn, M. Klawe, and D. Kleitman. Traditional galleries require fewer watchmen. *SIAM Journal on Algebraic and Discrete Methods*, 4(2), 194–206, 1983. https://doi.org/10.1137/0604020

[17] M. J. Katz and G. S. Roisman. On guarding the vertices of rectilinear domains. *Computational Geometry: Theory and Applications*, 39(3), 219–228, 2008. https://doi.org/10.1016/j.comgeo.2007.02.002

[18] D. T. Lee and A. K. Lin. Computational complexity of art gallery problems. *IEEE Transactions on Information Theory*, 32(2), 276–282, 1986. https://doi.org/10.1109/TIT.1986.1057165

[19] J. Maßberg. Perfect graphs and guarding rectilinear art galleries. *Discrete & Computational Geometry*, 51, 569–577, 2014. https://doi.org/10.1007/s00454-014-9587-4

[20] J. Maßberg. *Geometrical and Combinatorial Optimization Problems*. Habilitationsschrift, Universität Ulm, 2015. https://www.uni-ulm.de/fileadmin/website_uni_ulm/mawi.inst.080/Massberg/habilitation_massberg.pdf

[21] T. S. Michael and V. Pinciu. How to guard orthogonal polygons: diagonal graphs and vertex covers. *Discrete & Computational Geometry*, 55(2), 410–422, 2016. https://doi.org/10.1007/s00454-015-9756-0

[22] R. Motwani, A. Raghunathan, and H. Saran. Covering orthogonal polygons with star polygons: The perfect graph approach. *Journal of Computer and System Sciences*, 40(1), 19–48, 1990. https://doi.org/10.1016/0022-0000(90)90017-F

[23] J. O'Rourke. *Art Gallery Theorems and Algorithms*. Oxford University Press, 1987. https://www.science.smith.edu/~jorourke/books/ArtGalleryTheorems/Art_Gallery_Full_Book.pdf

[24] M. S. Paterson and F. F. Yao. Optimal binary space partitions for orthogonal objects. *Journal of Algorithms*, 13(1), 99–113, 1992. https://doi.org/10.1016/0196-6774(92)90007-Y

[25] V. Pinciu. Guarding polyominoes, polycubes and polyhypercubes. *Electronic Notes in Discrete Mathematics*, 49, 159–166, 2015. https://doi.org/10.1016/j.endm.2015.06.024 Open FWCG version: https://cse.buffalo.edu/fwcg2015/assets/pdf/FWCG_2015_paper_14.pdf

[26] D. Schuchardt and H.-D. Hecker. Two NP-hard art-gallery problems for orthogonal polygons. *Mathematical Logic Quarterly*, 41(2), 261–267, 1995. https://doi.org/10.1002/malq.19950410212

[27] T. C. Shermer. Recent results in art galleries. *Proceedings of the IEEE*, 80(9), 1384–1399, 1992. https://doi.org/10.1109/5.163407

[28] J. Urrutia. Art gallery and illumination problems. In J.-R. Sack and J. Urrutia, editors, *Handbook of Computational Geometry*, 973–1027. Elsevier, 2000. https://doi.org/10.1016/B978-044482537-7/50023-1

[29] J. Urrutia. Illumination of polygons: open problems. https://www.matem.unam.mx/~urrutia/openprob/Polygons/

[30] G. Viglietta. Guarding and searching polyhedra. Ph.D. thesis; arXiv:1211.2483, 2012. https://arxiv.org/abs/1211.2483 https://giovanniviglietta.com/papers/thesis.pdf

[31] G. Viglietta. Face-guarding polyhedra. *Computational Geometry: Theory and Applications*, 47(8), 833–846, 2014. https://doi.org/10.1016/j.comgeo.2014.04.009

[32] G. Viglietta. Optimally guarding 2-reflex orthogonal polyhedra by reflex edge guards. *Computational Geometry: Theory and Applications*, 86, 101589, 2020. https://doi.org/10.1016/j.comgeo.2019.101589 https://arxiv.org/abs/1708.05469

[33] C. Worman and J. M. Keil. Polygon decomposition and the orthogonal art gallery problem. *International Journal of Computational Geometry & Applications*, 17(2), 105–138, 2007. https://doi.org/10.1142/S0218195907002264

[34] P. Zylinski. Orthogonal art galleries with holes: A coloring proof of Aggarwal's theorem. *The Electronic Journal of Combinatorics*, 13(1), R20, 2006. https://doi.org/10.37236/1046 https://www.combinatorics.org/ojs/index.php/eljc/article/view/v13i1r20